

26.3 Condition indices and variance decomposition

A variance inflation factor says how much one coefficient has suffered, but not which other regressors are responsible, how many near-dependencies there are, or whether the intercept is involved. The eigenvalue analysis of [Section 26.1](#) answers those questions but depends on the units of the columns. This section fixes a scaling and derives the variance-decomposition proportions of Belsley, Kuh and Welsch (1980).

26.3.1 Scaling the columns

The canonical directions of $\mathbf{X}^\top \mathbf{X}$ change when a column is rescaled (Exercise (A3)), and a merely short column produces a small eigenvalue without any near-dependence (Exercise (B1)). The diagnostics are therefore computed for the design with every column, including $\mathbf{1}$, scaled to unit length:

$$\tilde{\mathbf{X}} = \mathbf{X}\mathbf{D}_0, \quad \mathbf{D}_0 = \text{diag}(1/\|\mathbf{x}_1\|, \dots, 1/\|\mathbf{x}_p\|).$$

By Proposition 10.5.2 this is within a factor $\sqrt{p}$ of the best diagonal scaling for the condition number. The coefficients of $\tilde{\mathbf{X}}$ are $\tilde{\beta}_j = \|\mathbf{x}_j\|\beta_j$.

Definition 26.3.1 — Condition indices

Let $\tilde{\mathbf{X}}$ have singular values $\mu_1 \geq \mu_2 \geq \dots \geq \mu_p > 0$. The **condition indices** of $\mathbf{X}$ are

$$\eta_k = \frac{\mu_1}{\mu_k}, \quad k = 1, \dots, p.$$

The largest, $\eta_p = \kappa(\tilde{\mathbf{X}})$, is the **scaled condition number**.

Whether to centre before scaling is a real choice. Let $\mathbf{R}$ be the correlation matrix of the non-intercept columns, the cross-product matrix of the centred and scaled design, and write $s_j^2 = S_{jj}/n$ and $\text{CV}_j = s_j/|\bar{x}_j|$ for the **coefficient of variation** of a column with nonzero mean.

Proposition 26.3.1 — Scaled condition numbers

Let $\mathbf{X}$ contain the column $\mathbf{1}$ and have full column rank, and let $\mathbf{R}$ be nonsingular.

- (a) $\sum_k \mu_k^2 = p$, so $1 \leq \mu_1^2 \leq p$. Every diagonal entry of $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$ is at most $1/\mu_p^2 \leq \eta_p^2$.
- (b) $\lambda_{\min}(\mathbf{R}) \geq \mu_p^2$. Hence $\eta_p^2 \geq 1/\lambda_{\min}(\mathbf{R}) \geq \max_j \text{VIF}_j$: every near-dependency among the centred regressors is at least as severe in the uncentred scaled design.
- (c) If a non-intercept column $\mathbf{x}_j$ has $\bar{x}_j \neq 0$, then $\mu_p^2 \leq \text{CV}_j^2/(\text{CV}_j^2 + 2)$, so that $\eta_p \geq \sqrt{(\text{CV}_j^2 + 2)/\text{CV}_j^2} > \sqrt{2}/\text{CV}_j$.

Proof.

- (a) $\sum_k \mu_k^2 = \|\tilde{\mathbf{X}}\|_F^2 = p$ (Proposition 1.8.2), since each column has unit length. The

largest of p numbers with sum p is between 1 and p . The j th diagonal entry of $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$ is a Rayleigh quotient of a matrix whose largest eigenvalue is $1/\mu_p^2$ (Theorem 1.7.7), and $1/\mu_p^2 = \eta_p^2/\mu_1^2 \leq \eta_p^2$.

- (b) Number the columns so that $\mathbf{x}_0 = \mathbf{1}$ and $\mathbf{x}_1, \dots, \mathbf{x}_k$ are the others. The centred and scaled columns are $\mathbf{z}_j = (\mathbf{x}_j - \bar{x}_j \mathbf{1})/(\sqrt{n} s_j)$, so that $\mathbf{R} = \mathbf{Z}^\top \mathbf{Z}$ with $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_k]$. Since $\mathbf{1} = \sqrt{n} \tilde{\mathbf{x}}_0$ and $\mathbf{x}_j = \|\mathbf{x}_j\| \tilde{\mathbf{x}}_j$, where $\tilde{\mathbf{x}}_j$ are the columns of $\tilde{\mathbf{X}}$,

$$\mathbf{Z}\mathbf{c} = \sum_{j=1}^k c_j \frac{\|\mathbf{x}_j\| \tilde{\mathbf{x}}_j - \sqrt{n} \bar{x}_j \tilde{\mathbf{x}}_0}{\sqrt{n} s_j} = \tilde{\mathbf{X}}\mathbf{b},$$

where

$$b_j = \frac{c_j \|\mathbf{x}_j\|}{\sqrt{n} s_j} \quad (j \geq 1), \quad b_0 = -\sum_{j=1}^k \frac{c_j \bar{x}_j}{s_j}.$$

Because $\|\mathbf{x}_j\|^2 = n(s_j^2 + \bar{x}_j^2) \geq ns_j^2$, $|b_j| \geq |c_j|$, so $\|\mathbf{b}\| \geq \|\mathbf{c}\|$. For a unit vector $\mathbf{c}$, $\mathbf{c}^\top \mathbf{R} \mathbf{c} = \|\tilde{\mathbf{X}}\mathbf{b}\|^2 \geq \mu_p^2 \|\mathbf{b}\|^2 \geq \mu_p^2$ by Theorem 1.7.7; minimizing over $\mathbf{c}$ gives $\lambda_{\min}(\mathbf{R}) \geq \mu_p^2$. Then $\eta_p^2 \geq 1/\mu_p^2 \geq 1/\lambda_{\min}(\mathbf{R})$ by (a), and $1/\lambda_{\min}(\mathbf{R}) \geq \text{VIF}_j$ by Proposition 19.6.1(c).

- (c) Take $\mathbf{b}$ with $b_j = \|\mathbf{x}_j\|$, $b_0 = -\sqrt{n} \bar{x}_j$ and all other entries zero. Then $\tilde{\mathbf{X}}\mathbf{b} = \mathbf{x}_j - \bar{x}_j \mathbf{1}$, with squared length ns_j^2 , while $\|\mathbf{b}\|^2 = n(s_j^2 + \bar{x}_j^2) + n\bar{x}_j^2$. So $\mu_p^2 \leq ns_j^2/(n(s_j^2 + 2\bar{x}_j^2)) = \text{CV}_j^2/(\text{CV}_j^2 + 2)$, and $\eta_p \geq 1/\mu_p$ by (a).

■

Part (b) is the formal content of the argument of Belsley (1984) for not centring: centring can only hide near-dependencies, never create them. Part (c) shows what it hides. A regressor that varies little relative to its mean is nearly a multiple of the column of ones, and that near-dependency with the intercept is invisible in $\mathbf{R}$ and in every variance inflation factor. The design of Example (A macroeconomic design) shows the size of the effect: scaled to unit length, its condition number is 383; centred and scaled, 124.

Whether the hidden dependency matters depends on the question, as in Example (A quadratic far from the origin). It inflates the variance of the intercept, the mean response at the origin, and of the coefficients whose columns take part in it. If the origin is far from the data, it concerns a question nobody asked; if the origin is meaningful, the uncentred diagnostic is the honest one. This chapter uses the uncentred scaled design, as Belsley, Kuh and Welsch do, and reports the centred condition number alongside.

26.3.2 The variance-decomposition proportions

Let the singular value decomposition of the scaled design be

$$\tilde{\mathbf{X}} = \sum_{k=1}^p \mu_k \mathbf{u}_k \mathbf{t}_k^\top,$$

with orthonormal right singular vectors $\mathbf{t}_k = (t_{1k}, \dots, t_{pk})^\top$. The $p \times p$ matrix with columns $\mathbf{t}_k$ is orthogonal, so its rows also have unit length: $\sum_k t_{jk}^2 = 1$ for every j .

Theorem 26.3.2 — Variance-decomposition proportions

Let $\mathbf{X}$ have rank p , $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, and let $\phi_{jk} = t_{jk}^2 / \mu_k^2$.

- (a) $\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|\mathbf{x}_j\|^2} \sum_{k=1}^p \phi_{jk}$. The **variance-decomposition proportions**

$$\pi_{kj} = \frac{\phi_{jk}}{\sum_{l=1}^p \phi_{jl}}$$

satisfy $\sum_k \pi_{kj} = 1$, do not depend on σ^2 , and vanish iff $t_{jk} = 0$. The term π_{kj} is the share of $\text{Var}(\hat{\beta}_j)$ contributed by the canonical coordinate with condition index η_k .

- (b) For each k , $\|\sum_j t_{jk} \tilde{\mathbf{x}}_j\| = \mu_k$ while $\|\mathbf{t}_k\| = 1$: the vector $\mathbf{t}_k$ holds the coefficients of a near-dependency among the scaled columns, of strength μ_k .
- (c) Let S be a nonempty proper subset of $\{1, \dots, p\}$ such that $\mu_k \leq \mu_S$ for $k \in S$ and $\mu_k \geq \mu_L$ for $k \notin S$, with $\mu_S < \mu_L$. Let $w_j = \sum_{k \in S} t_{jk}^2$ measure the participation of column j in the near-dependencies indexed by S . Then

$$\sum_{k \in S} \pi_{kj} \geq \frac{w_j}{w_j + (1 - w_j) \mu_S^2 / \mu_L^2}, \quad \text{Var}(\hat{\beta}_j) \geq \frac{\sigma^2 w_j}{\|\mathbf{x}_j\|^2 \mu_S^2}.$$

Proof.

- (a) Since $\mathbf{X}\boldsymbol{\beta} = \tilde{\mathbf{X}}\mathbf{D}_0^{-1}\boldsymbol{\beta}$, the least squares coefficients of $\tilde{\mathbf{X}}$ are $\hat{\tilde{\boldsymbol{\beta}}} = \|\mathbf{x}_j\| \hat{\beta}_j$, and by Theorem 26.1.1(a) applied to $\tilde{\mathbf{X}}$, whose cross-product matrix has eigenvalues μ_k^2 and eigenvectors $\mathbf{t}_k$, $\text{Cov}(\hat{\tilde{\boldsymbol{\beta}}}) = \sigma^2 \sum_k \mathbf{t}_k \mathbf{t}_k^\top / \mu_k^2$. Its j th diagonal entry is $\sigma^2 \sum_k \phi_{jk}$. The statements about the proportions follow from the definition, since $\mu_k > 0$.
- (b) $\sum_j t_{jk} \tilde{\mathbf{x}}_j = \tilde{\mathbf{X}} \mathbf{t}_k = \mu_k \mathbf{u}_k$.
- (c) Put $A = \sum_{k \in S} \phi_{jk}$ and $B = \sum_{k \notin S} \phi_{jk}$. Then $A \geq w_j / \mu_S^2$ and, since $\sum_{k \notin S} t_{jk}^2 = 1 - w_j$, $B \leq (1 - w_j) / \mu_L^2$. The sum of the proportions over S is $A / (A + B)$, which increases with A and decreases with B , so it is at least $(w_j / \mu_S^2) / (w_j / \mu_S^2 + (1 - w_j) / \mu_L^2)$, which is the stated bound. The variance bound is $\text{Var}(\hat{\beta}_j) \geq \sigma^2 A / \|\mathbf{x}_j\|^2$ together with $A \geq w_j / \mu_S^2$. ■

Part (c) is why the proportions locate near-dependencies. If the condition indices in S are at least ten times those outside it, $\mu_S^2 / \mu_L^2 \leq 0.01$, then any column with even a tenth of its unit row inside S , $w_j \geq 0.1$, has at least $0.1 / (0.1 + 0.9 \times 0.01) \approx 0.92$ of its variance attributed to S , and its variance is inflated accordingly. A column that does not take part has proportions near zero there. When several small singular values are close together, the individual $\mathbf{t}_k$ are poorly determined, and a column's proportions can be split between their rows in unpredictable ratios. Part (c) bounds the *sum* over the group, which is stable, and that is what should be read ([Exercise \(C1\)](#)).

Idea. Belsley, Kuh and Welsch's reading of the table: scale the columns to unit length; flag condition indices of about 30 or more (indices near 10 signal weak dependencies); in each flagged row, or group of rows with similar indices, look for two or more coef-

ficients with large proportions, conventionally above 0.5. Those columns form a near-dependency. Confirm it by regressing one of them on the others.

Example. (A consumption function)

Return to the regression of Example (Large factors, precise estimate), with an intercept and six regressors. The scaled design has condition indices and proportions

Index	const.	income	pop.	CPI	M1	T-bill	unemp.
1.0	0.000	0.000	0.000	0.000	0.000	0.001	0.001
3.4	0.000	0.000	0.000	0.000	0.001	0.053	0.007
7.7	0.000	0.000	0.000	0.001	0.001	0.347	0.090
13.4	0.002	0.001	0.000	0.001	0.002	0.001	0.688
38.9	0.005	0.149	0.000	0.001	0.290	0.049	0.036
74.3	0.007	0.130	0.000	0.995	0.569	0.241	0.116
270.9	0.985	0.720	0.999	0.001	0.138	0.307	0.060

Figure 26.3.1 shows the table graphically. Three indices exceed 30, and the table separates them.

- *Index 270.9: intercept, income and population* (proportions 0.985, 0.720, 0.999). The singular vector has entries -0.522 , -0.236 and 0.817 on these columns and small ones elsewhere: population, which grew steadily with income, is nearly an affine function of it ($R^2 = 0.9885$). Its small coefficient of variation, 0.156, alone forces an index of at least 9.1 by Proposition 26.3.1(c); the shared trend does the rest.
- *Index 74.3: the price level and money*. CPI has proportion 0.995 and M1 0.569. Regressing CPI on income and M1 gives $R^2 = 0.9868$.
- *Index 38.9: a weaker dependency involving M1 and income*. Taken with the previous row, M1 has $0.290 + 0.569 = 0.859$ of its variance in the two rows, and income has $0.149 + 0.130 + 0.720 = 0.999$ in the three rows above 30. Income takes part in all three near-dependencies.

A proportion is a share, not a size: the 0.307 of the T-bill rate at index 270.9 is a share of a variance that collinearity has barely inflated (its variance inflation factor is 2.9). Centred and scaled, the design has condition number 37.9 against 270.9 uncentred: centring removes the intercept from the largest near-dependency, and what is left is the relation between income and population that the variance inflation factors 154.5 and 183.7 already showed.

```
import numpy as np
import statsmodels.api as sm
macro = sm.datasets.macrodta.load_pandas().data
names = ["const", "realdpi", "pop", "cpi", "m1", "tbilrate", "unemp"]
X = sm.add_constant(macro[names[1:]].to_numpy())
X_s = X / np.linalg.norm(X, axis=0)                                # unit-length columns, not
    centred                                                         # condition indices

U, mu, Vt = np.linalg.svd(X_s, full_matrices=False)              # X_s = U diag(mu) V'
eta = mu[0] / mu
```

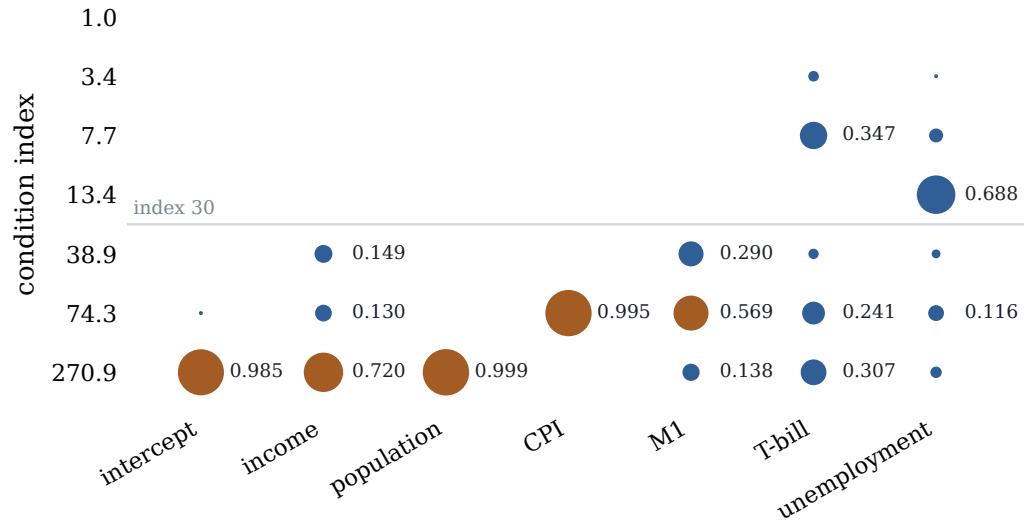


Figure 26.3.1: Variance-decomposition proportions of [Example \(A consumption function\)](#): rows are condition indices, columns coefficients, dot area proportional to π_{kj} . Highlighted: $\pi_{kj} \geq 0.5$ with index at least 30.

```
phi = Vt.T ** 2 / mu**2                                # phi[j, k] = v_jk^2 / mu_k^2
prop = (phi / phi.sum(axis=1, keepdims=True)).T         # row k: proportions for index
eta_k

print("index  " + " ".join(f"{nm[:7]:>7s}" for nm in names))
for k in range(len(mu)):
    print(f"{eta[k]:7.1f}" + " ".join(f"{p:7.3f}" for p in prop[k]))
```

[Run this code online](#)

```
Z = X[:, 1:] - X[:, 1:].mean(axis=0)
Z_s = Z / np.linalg.norm(Z, axis=0)                    # centred, then unit length
mu_c = np.linalg.svd(Z_s, compute_uv=False)
print("scaled condition number, not centred:", round(eta[-1], 1))
print("scaled condition number, centred    :", round(mu_c[0] / mu_c[-1], 1))
cv = X[:, 1:].std(axis=0) / X[:, 1:].mean(axis=0)
print("coefficients of variation:", dict(zip(names[1:], np.round(cv, 3))))
```

[Run this code online](#)

```
def r_squared(target, others):
    fit = sm.OLS(macro[target], sm.add_constant(macro[others])).fit()
    return fit.rsquared

print("pop on realdpi      : R^2 =", round(r_squared("pop", ["realdpi"]), 4))
print("cpi on realdpi, m1 : R^2 =", round(r_squared("cpi", ["realdpi", "m1"]), 4))
```

[Run this code online](#)

The script behind the example checks the bound of [Theorem 26.3.2\(c\)](#) for the groups formed by the last one, two and three indices, and the inequalities of [Proposition 26.3.1](#).

26.3.3 Single cases

Collinearity diagnostics can be dominated by a few cases: one far out along a near-dependency can create it, one lying across it can hide it. The deletion formulas of [Chapter 20](#) show that only cases of high leverage can have much effect.

Proposition 26.3.3 — Deleting a case

Let $\mathbf{X}$ have rank p , let $\mathbf{x}_{(i)}$ be its i th row with leverage $h_{ii} < 1$, let $\mathbf{X}_{[i]}$ be $\mathbf{X}$ without that row (Theorem 20.3.1 writes $\mathbf{X}_{(i)}$; square brackets here avoid a clash with $\mathbf{X}_{(j)}$, the matrix without column j), and write $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^{-1}$, $\mathbf{G}_{[i]} = (\mathbf{X}_{[i]}^\top \mathbf{X}_{[i]})^{-1}$.

- (a) $\mathbf{a}^\top \mathbf{G}_{[i]} \mathbf{a} = \mathbf{a}^\top \mathbf{G} \mathbf{a} + (\mathbf{a}^\top \mathbf{G} \mathbf{x}_{(i)})^2 / (1 - h_{ii})$ for every $\mathbf{a}$.
- (b) $\mathbf{a}^\top \mathbf{G} \mathbf{a} \leq \mathbf{a}^\top \mathbf{G}_{[i]} \mathbf{a} \leq \mathbf{a}^\top \mathbf{G} \mathbf{a} / (1 - h_{ii})$: deleting case i multiplies the variance of every linear function of $\hat{\boldsymbol{\beta}}$ by a factor between 1 and $1/(1 - h_{ii})$.
- (c) $(1 - h_{ii}) \lambda_{\min}(\mathbf{X}^\top \mathbf{X}) \leq \lambda_{\min}(\mathbf{X}_{[i]}^\top \mathbf{X}_{[i]}) \leq \lambda_{\min}(\mathbf{X}^\top \mathbf{X})$.

Proof.

- (a) This is the rank-one deletion formula $\mathbf{G}_{[i]} = \mathbf{G} + \mathbf{G} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \mathbf{G} / (1 - h_{ii})$ of Example (Deleting an observation), on which Theorem 20.3.1 rests, sandwiched between $\mathbf{a}^\top$ and $\mathbf{a}$.
- (b) The lower bound is clear from (a). For the upper, apply the Cauchy–Schwarz inequality to the inner product $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{G} \mathbf{b}$, which is legitimate because $\mathbf{G}$ is positive definite: $(\mathbf{a}^\top \mathbf{G} \mathbf{x}_{(i)})^2 \leq (\mathbf{a}^\top \mathbf{G} \mathbf{a}) h_{ii}$. Then $\mathbf{a}^\top \mathbf{G}_{[i]} \mathbf{a} \leq \mathbf{a}^\top \mathbf{G} \mathbf{a} (1 + h_{ii} / (1 - h_{ii}))$.
- (c) For a positive definite matrix, $1/\lambda_{\min}$ is the largest eigenvalue of the inverse, which is $\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top (\cdot)^{-1} \mathbf{a}$ (Theorem 1.7.7). Taking the maximum in (b) over unit vectors gives the two inequalities. ■

So removing a case of small leverage changes the smallest eigenvalue by at most the factor $1 - h_{ii}$, and no variance by more than $1/(1 - h_{ii})$. In the consumption regression the largest leverage is 0.161, at 2009Q3, so by (c) no single quarter can reduce the smallest eigenvalue of the scaled cross-product matrix by more than the factor 1.192, with the column scaling of the full data held fixed. The largest reduction actually achieved by deleting one of the 203 quarters is by the factor 1.055. The near-dependencies of this design are carried by the whole sample, not by a few quarters.

Adding a case never increases a variance, by (b) read the other way round, but it can increase every variance inflation factor and the condition number ([Exercise \(B2\)](#)).

26.3.4 Exercises

26.3.4.1 A. Check your understanding

Exercise A1.

In the table of [Example \(A consumption function\)](#), why is the intercept's variance almost entirely attributed to the largest index, although the intercept is not an economic variable? Which coefficients would you expect to change most if a few more years of data were added in which population grew faster than income?

26.3.4.2 B. Practice**Exercise B1.**

Let $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\bar{x} \neq 0$. Show that the scaled design has $\mu_{1,2}^2 = 1 \pm \cos \theta$, where $\cos \theta = 1/\sqrt{1 + CV^2}$ is the cosine of the angle between $\mathbf{1}$ and $\mathbf{x}$, and hence $\eta_2^2 = (1 + \cos \theta)/(1 - \cos \theta)$. Compare with the bound of [Proposition 26.3.1\(c\)](#). What is the variance inflation factor of $\mathbf{x}$?

Solution. The scaled columns are unit vectors at angle θ , so $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}$ with $c = \cos \theta$, whose eigenvalues are $1 \pm c$ (Exercise (B1)). Now $c = \mathbf{1}^\top \mathbf{x} / (\sqrt{n} \|\mathbf{x}\|) = n\bar{x} / \sqrt{n \cdot n(s^2 + \bar{x}^2)} = 1/\sqrt{1 + CV^2}$ (taking $\bar{x} > 0$). So $\eta_2^2 = (1 + c)/(1 - c)$. The bound of the proposition gives $\mu_2^2 \leq CV^2/(CV^2 + 2)$. For small CV , $1 - c \approx CV^2/2$, so $\mu_2^2 = 1 - c \approx CV^2/2$, and the bound is attained to first order. With a single regressor $R_j^2 = 0$ and $VIF = 1$: the variance inflation factor cannot see this near-dependency at all.

Exercise B2.

Take $n = 20$ cases with (x_1, x_2) at the points of a 4×5 grid in $[-1, 1]^2$, so that the regressors are uncorrelated, and add a twenty-first case at (t, t) . Show that the correlation of the two regressors tends to one as $t \rightarrow \infty$, so that both variance inflation factors grow without bound, while no variance of any linear function of $\hat{\beta}$ increases. Reconcile the two statements.

Solution. By [Proposition 26.3.3\(b\)](#) read backwards, adding a case can only decrease $\mathbf{a}^\top \mathbf{G} \mathbf{a}$ for every $\mathbf{a}$. The sample covariance of x_1, x_2 in the enlarged data is $t^2(1 - 1/21)/21$ (the grid contributes nothing, being centred and uncorrelated), and the sample variance of x_j is $(S_j + t^2(1 - 1/21))/21$, where S_j is the grid's sum of squares for x_j . As $t \rightarrow \infty$ the correlation tends to one and $VIF_j = 1/(1 - r^2) \rightarrow \infty$. But $\text{Var}(\hat{\beta}_j) = \sigma^2 VIF_j / S_{jj}$, and S_{jj} grows like t^2 as fast as VIF_j does, so the variances converge to finite limits no larger than before. The variance inflation factor compares with an orthogonal design with the *new* spread, which the added point has made enormous; the point added information along $(1, 1)$ and none across it.

26.3.4.3 C. Going deeper**Exercise C1.**

Suppose $\mu_k = \mu$ for all $k \in S$. Show that the singular vectors $\mathbf{t}_k$, $k \in S$, are determined

only up to an orthogonal transformation of their span, that the individual proportions π_{kj} , $k \in S$, depend on the choice, and that $\sum_{k \in S} \pi_{kj}$ does not.

Solution. If $\mathbf{T}_S$ has columns $\mathbf{t}_k$, $k \in S$, and $\mathbf{O}$ is orthogonal, then $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{T}_S \mathbf{O} = \mu^2 \mathbf{T}_S \mathbf{O}$, so the columns of $\mathbf{T}_S \mathbf{O}$ are equally valid singular vectors, and the individual t_{jk}^2 change with $\mathbf{O}$. But $\sum_{k \in S} \phi_{jk} = \mu^{-2} \sum_{k \in S} t_{jk}^2 = \mu^{-2} \mathbf{e}_j^\top \mathbf{T}_S \mathbf{T}_S^\top \mathbf{e}_j$, and $\mathbf{T}_S \mathbf{T}_S^\top$ is the projection onto the span, which does not depend on $\mathbf{O}$. The denominator $\sum_l \phi_{jl}$ is a diagonal entry of $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$ and does not depend on the choice either.